

Bibek Neupane

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EDUCATION

Fisk University

Bachelor of Science in Computer Science | GPA: 3.80

Expected Graduation: Dec 2026

Nashville, TN

- **Relevant Courses:** Database Management, Data Structures & Algorithms, Operating Systems, Natural Language Processing (NLP), Theory of Computation, Object Oriented Programming (OOP), Programming Languages, Machine Learning, Linear Algebra, Probability and Statistics, Calculus I & II
- **Awards:** Fisk University Outstanding Scholars and Leaders Scholar, Goldman Sachs Market Madness (HBCU Possibilities) 2025, Emerging Research Scholar

TECHNICAL SKILLS

- **Languages:** Python, C/C++, JavaScript, TypeScript, Java, C#, SQL, HTML5/CSS3
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis
- **Tools & Frameworks:** ReactJS, NodeJS, ExpressJS, Django, Flask, PyTorch, TensorFlow, Tableau, Git/GitHub, GitHub Actions, Docker, AWS (EC2), Linux, Jest, Firebase, .NET

EXPERIENCE

TN-QuMat | *AMPLIFY Quantum Research Scholar*

Jul 2026 – Sep 2026

- Investigating **quantum threats to modern cryptography** by implementing Shor's and Grover's algorithms in Qiskit and analyzing their impact on RSA, ECC, and symmetric encryption.
- Implementing and benchmarking NIST post-quantum standards **ML-KEM** and **ML-DSA**, comparing key, ciphertext, signature sizes, and performance against classical RSA/ECC for quantum-resistant secure communication.

NexSewa | *Software Engineering Intern*

May 2025 – Aug 2025

- Built a full-stack web app using **React**, **Node.js/Express**, and **PostgreSQL** for internal support staff to manage customer requests; optimizing APIs and queries to reduce response time by **35%**.
- Developed an AI chatbot and task agent using **OpenAI APIs** to automate customer support and scheduling workflows, saving **25+ hours/month**.
- Implemented **WebSockets** and **Redis** caching, set up CI/CD with **GitHub Actions**, wrote **Jest** tests (**90%+** coverage), and deployed on **AWS EC2** using **Docker**.

Mastercard | *Team UnderCoder — Mastercard X AUC Data Science (First Runner-up)*

Oct 2025 – Dec 2025

- Built a **Random Forest** model in **scikit-learn** to predict Inclusive Growth Scores for underserved Tennessee community
- Created a **Python** pipeline using **Pandas** and **NumPy** to process datasets and engineer features.
- Developed visualizations using **Matplotlib** and **Plotly** to communicate community disparities and trends.

Chain-Link Engineering | *Software Engineering Intern*

May 2024 – Aug 2024

- Contributed to a full-stack web app, an online hospital ticketing system, using **React.js** and **Node.js/Express**, designing and consuming RESTful APIs to streamline patient check-in workflows for **700+** daily users.
- Designed **MongoDB** schemas and built data dashboards powered by backend APIs, enabling hospital management to access real-time insights and improving decision speed by **15%**.

PROJECTS

MiniDB: SQL Database Engine | [GitHub](#) | *Go, SQL Parser, File-Based Storage, Unit Testing*

- Built a SQL database engine in **Go** with a custom **REPL**, parser, executor, catalog, and JSON-lines storage layer.
- Supported CREATE, INSERT, SELECT, UPDATE, DELETE, and ALTER TABLE without external databases.
- Added disk persistence, hidden row IDs, primary key validation, and unit tests for parser, storage, and executor packages.

Financial Sentiment Analysis (BERT) | [GitHub](#) | *Python, PyTorch, Hugging Face Transformers, Streamlit*

- Trained and fine-tuned a custom **BERT** ('bert-base-uncased') model on financial text for 3-class sentiment classification: **negative**, **neutral**, and **positive**.
- Built preprocessing, normalization, and tokenization pipeline for structured training and validation.
- Built **Streamlit** apps for real-time inference and side-by-side comparison of the custom model against **ProsusAI/FinBERT** with probability-based outputs.

Soccer-Stat Analyzer | [GitHub](#) | *ReactJS, MongoDB, FootballAPI, ExpressJS, NodeJS*

- Developed a React-based application to track global soccer clubs, providing real-time data on fixtures, results, and player statistics via **API-Football**.
- Added **MongoDB** caching to reduce external API calls and improve latency during repeated requests.
- Developed **Node.js/Express.js** backend routes for real-time data retrieval and server-side processing.

ACTIVITIES AND HACKATHONS

Google Developers Students Club, Fisk CS Club, VandyHacks XI, Be Smart Hackathon, Google HBCU Hack